

MATHEMATICAL MODELS OF ELECTROMAGNETIC WAVE SCATTERING BY TWO-ELEMENT STRIP GRATING WITH A PERPENDICULARLY MAGNETIZED GYROTROPIC MEDIUM

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ABSTRACT

Two mathematical models are proposed for analyzing a linearly polarized plane wave scattering by a strip grating placed on isotropic-gyrotropic media interface in the case of oblique incidence. The first mathematical model is based on reducing the original boundary value problem to the Riemann-Hilbert problem. The second model is based on reducing the same problem to a singular integral equation of the first kind with Cauchy kernel and its numerical solving by the discrete singularities method.

In this paper, the results obtained in [1-3] are generalized for the case of periodic structure consisting of two strips of different widths per period (two-element grating). This results in richer diffraction phenomena in comparison to simple grating because of additional control parameters. Moreover, unlike papers [1-3] here the case of oblique incidence of an H-polarized plane wave on a two-element grating is considered. The center of coordinate system is chosen in the middle of one of the strips.

The following set of dual series equations is mathematical model of a structure

$$\sum_n A_n \gamma_{n1} e^{in\varphi} = \gamma_{01}, \quad \theta_m < \varphi < \theta_m^{(1)}; \quad (1)$$

$$\sum_n A_n r_n e^{in\varphi} = -p_0, \quad \theta_m^{(1)} < \varphi < \theta_{m+1}^{(1)}, \quad (2)$$

where $\gamma_{n1} = \sqrt{k_0^2 n_1^2 - h_n^2}$, $n_1 = \sqrt{\varepsilon_1 \mu_1}$, $h_n = k_0 n_1 \sin \zeta + \frac{2\pi}{l} n$,

$$r_n = 1 + \frac{\gamma_{n1}}{\varepsilon_1 (R \gamma_{n2} - i L h_n)}, \quad \gamma_{n2} = \sqrt{k_0^2 \varepsilon_{\perp} \mu_{\parallel} - h_n^2}, \quad \varepsilon_{\perp} = \frac{\varepsilon^2 - \varepsilon_a^2}{\varepsilon}$$

$$R = \varepsilon_{\perp}^{-1}, \quad L = -\frac{\varepsilon_a}{\varepsilon^2 - \varepsilon_a^2}, \quad k_0 = \frac{2\pi}{\lambda},$$

$$p_0 = 1 - \frac{\gamma_{01}}{\varepsilon_1 (R \gamma_{02} - i L h_0)}, \quad \varphi = \frac{2\pi}{l} y, \quad \theta_m = 2\pi \frac{y_m}{l}, \quad (m=1,2),$$

l is the grating period, d is the slot width, λ is the wavelength, ζ is the incidence angle.

Denote $\tilde{A}_n = A_n r_n + p_0 \delta_{0n}$, and δ_{0n} for the Kronecker symbol.

Then initial set of dual series equations takes the form:

$$\sum_n \tilde{A}_n e^{in\varphi} = 0, \quad \theta_m < \varphi < \theta_m^{(1)}; \quad (3)$$

$$\sum_{n>0} n \tilde{A}_n \frac{\eta_{n1}}{r_n^+} e^{in\varphi} + \sum_{n<0} |n| \tilde{A}_n \frac{\eta_{n1}}{r_n^-} e^{in\varphi} = (1 - \frac{\tilde{A}_0 - p_0}{r_0}) \kappa_1 \cos \zeta, \quad \theta_m^{(1)} < \varphi < \theta_{m+1}^{(1)}, \quad (4)$$

$$\sum_{n \neq 0} \tilde{A}_n e^{in\delta_m} = -\tilde{A}_0, \quad \theta_m^{(1)} < \delta_m < \theta_{m+1}^{(1)}, \quad (5)$$

The set of equations (3)-(5) can be reduced to non-homogeneous conjugation problem (Riemann-Hilbert's problem) with a complex-valued coefficient in the case of account of dissipative losses.

To calculate matrix elements of final matrix equation, it is necessary to introduce polynomials $Q_n(u_m, \rho)$ [3], where $\rho = \frac{\ln|G|}{2\pi}$, $u_m = \cos\theta_m$, ($m=1,2$).

In the case of multi-element gratings, an efficient numerical-analytical method for solving these dual series equations was suggested in [4]. The method consists in reducing them to a singular integral equation of the first kind with the Cauchy kernel on the set of segments, and its following solution by the method of discrete singularities [4,5]. Integral equation is of the following form

$$\frac{1}{\pi} \int_L \frac{F(\xi)}{\xi - x} d\xi + \frac{1}{\pi} \int_L K(x, \xi) F(\xi) d\xi = f(x), \quad x \in L \quad (6)$$

where $L = \bigcup_{q=1}^m (a_q, b_q)$, $-\infty < a_1 < b_1 < \dots < a_m < b_m < +\infty$;

$f(x)$, $x \in \bar{L}$; $K(x, \xi)$, $x \in \bar{L}$, $\xi \in \bar{L}$ are known smooth functions, and function

$F(\xi)$, $\xi \in L$ is sought in the functional class whose restriction on interval (a_q, b_q) :

$$F_q(\xi) = F(\xi), \quad a_q < \xi < b_q, \quad q = 1, \dots, m$$

can be represented in the form

$$F_q(\xi) = \frac{v_q(\xi)}{\sqrt{(\xi - a_q)(b_q - \xi)}}, \quad a_q < \xi < b_q,$$

where $v_q(\xi)$, $\xi \in [a_q, b_q]$ is a smooth function.

The sought function $F(\xi)$, $\xi \in L$ satisfies additional conditions, which in general case are of the following form:

$$\frac{1}{\pi} \int_L S_p(\xi) F(\xi) d\xi = C_p, \quad p = 1, \dots, m, \quad (7)$$

where $S_p(\xi)$, $\xi \in [a_p, b_p]$ is a known smooth function, and C_p is a known constant.

In conclusion we shall present the discrete mathematical model that is a set of linear algebraic equations for numerical solution of the integral equation (6) with additional condition (7).

Denote

$$t_i^n = \cos \frac{2i-1}{2n} \pi, \quad i = 1, \dots, n; \quad t_{0j}^n = \cos \frac{j}{n} \pi, \quad j = 1, \dots, n-1;$$

$$g_k(\tau) = \frac{b_k - a_k}{2}\tau + \frac{b_k + a_k}{2}; \quad \xi_{qi}^{n_q} = g_q(t_i^{n_q}), \quad i=1, \dots, n_q; q=1, \dots, m$$

$$\chi_{pj}^{n_p} = g_p(t_{0j}^{n_p}), \quad j=1, \dots, n_p-1; p=1, \dots, m$$

To calculate approximate values $\{v_{qn_q}(\xi)\}_{q=1}^m$ of the desired functions $v_q(\xi)$, $q=1, \dots, m$ in

principal points $\{t_i^{n_q}\}_{i=1}^{n_q}$, we have a set of linear algebraic equations

(where $R(x, \xi) = \frac{1}{\xi - x} + K(x, \xi)$)

$$\sum_{q=1}^m \sum_{i=1}^{n_q} R(\chi_{pj}^{n_p}, \xi_{qi}^{n_q}) v_{qn_q}(\xi_{qi}^{n_q}) \frac{1}{n_q} = f(\chi_{pj}^{n_p}), \quad j=1, \dots, n_p-1; p=1, \dots, m,$$

$$\sum_{i=1}^{n_p} S_p(\xi_{pi}^{n_p}) v_{pn_p}(\xi_{pi}^{n_p}) \frac{1}{n_p} = C_p, \quad (j=n_p), p=1, \dots, m$$

The values of the physical characteristic of scattered field,

$$H = \int_L H(\xi) F(\xi) d\xi = \sum_{q=1}^m \int_{a_q}^{b_q} H_q(\xi) v_q(\xi) \frac{d\xi}{\sqrt{(\xi - a_q)(b_q - \xi)}}.$$

are expressed in terms of the functions $v_q(\xi)$, $\xi \in [a_q, b_q]$, $q=1, \dots, m$,

where $H_q(\xi)$, $\xi \in [a_q, b_q]$ are known functions.

Approximate values of

$$H_{\bar{n}} = \sum_{q=1}^m \sum_{i=1}^{n_q} H_q(\xi_{qi}^{n_q}) v_{qn_q}(\xi_{qi}^{n_q}) \frac{1}{n_q}, \quad \bar{n} = (n_1, \dots, n_m)$$

are calculated in numerical experiments.

Obtained results can be applied in the design and elaboration of various devices containing periodic structures with ferrite substrates or in plasma.

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